Title : “Blind Joint CFO, Phase Noise, and IQ Imbalance Estimation in OFDM Receivers using Pilot-Aided and Data-Aided Hybrid Algorithms”

Technical terms:-

1. CFO (Carrier Frequency Offset):
It is the mismatch between transmitter and receiver frequencies due to imperfect oscillators. It causes rotation of the signal and breaks Orthogonal Frequency Division Multiplexing orthogonality, leading to interference. It is estimated using pilot signals and corrected by rotating the signal back.

2. Phase Noise:
It is the random fluctuation of the signal’s phase caused by hardware instability. It makes the constellation blurry and degrades detection accuracy. It is tracked over time and corrected using decision-directed (data-aided) methods.

3. IQ Imbalance:
It is the mismatch between the I (in-phase) and Q (quadrature) signal paths in the receiver. It distorts the constellation into an elliptical shape and reduces performance. It is estimated using signal statistics and compensated using scaling/rotation correction.

---

4. Blind Estimation:
It means estimating impairments without using known reference signals. The receiver relies only on received data patterns. It is useful to save bandwidth and improve efficiency.

5. Pilot-Aided Estimation:
It uses known symbols (pilots) inserted in the signal to estimate impairments. It gives accurate initial estimates. It is used for coarse synchronization.

6. Data-Aided Estimation:
It uses detected data symbols to refine the estimation further. It improves accuracy after initial correction. It works in a feedback manner but can propagate errors if detection is wrong.

7. Hybrid Algorithm:
It combines pilot-aided (for initial correction) and data-aided (for fine tuning). This gives both accuracy and efficiency. It mimics real wireless receiver design.

Project Overview

This project focuses on designing an advanced receiver for Orthogonal Frequency Division Multiplexing systems that can handle real-world RF imperfections. In practical wireless systems, signals get distorted due to hardware limitations such as frequency mismatch, phase instability, and imbalance in signal paths. The goal of this project is to accurately estimate and compensate these impairments so that the received signal can be correctly decoded. A hybrid approach

combining pilot-based and data-driven techniques is used to achieve reliable and efficient correction.

Problems in Existing Systems

In real communication systems, multiple impairments occur simultaneously:

- Carrier Frequency Offset (CFO): Causes rotation of the signal and destroys subcarrier orthogonality.
- Phase Noise: Introduces random phase variations, making the signal unstable.
- IQ Imbalance**:** Distorts the signal due to mismatch in receiver components.

Most traditional methods:

- Handle only one impairment at a time
- Depend heavily on pilot signals (extra overhead)
- Perform poorly when impairments are combined

This project solves the limitation of conventional receivers by:

Performing joint estimation of multiple RF impairments
Reducing dependence on pilots using blind/data-aided techniques
Improving accuracy using a hybrid approach (pilot + data)

In simple terms:
It builds a smarter receiver that can detect and correct multiple errors together, efficiently and accurately.

Working of the Project: The system works in the following steps:

1. **Signal Transmission:**
   Data is modulated and transmitted using OFDM.
2. **Impairment Addition:**
   The transmitted signal is affected by CFO, phase noise, and IQ imbalance.
3. **Receiver Processing:**
   The receiver first converts the signal back to frequency domain.
4. **Pilot-Aided Estimation (Coarse):**
   Known pilot symbols are used to get an initial estimate of impairments.
5. **Data-Aided Estimation (Fine):**
   Detected data is used to refine the estimates further.
6. **Compensation:**
   The estimated errors are removed from the signal.
7. **Detection:**
   Cleaned signal is demodulated to recover original data.

Why Each Component is Used:-

- **OFDM:** Efficient and widely used modulation technique in modern systems
- **CFO Correction:** Maintains subcarrier orthogonality
- **Phase Noise Tracking:** Stabilizes signal phase over time
- **IQ Imbalance Compensation:** Restores correct signal shape
- **Pilot-Aided Method:** Provides reliable initial estimation
- **Data-Aided Method:** Improves accuracy without extra bandwidth
- **Hybrid Approach:** Combines reliability + efficiency

Importance of the Project

- Reflects real-world wireless receiver design
- Relevant to standards like 5G NR
- Demonstrates both RF understanding and DSP skills
- Shows ability to handle complex, practical problems

MATLAB flow :-

Bits → QAM → OFDM Tx → Channel + RF Impairments →

OFDM Rx → Estimation → Compensation → Detection → BER/EVM

CODE:-

```
clc; clear; close all;


%% PARAMETERS

N = 64;                 % Number of subcarriers

cpLen = 16;             % Cyclic prefix length

numSymbols = 50;        % OFDM symbols

SNR = 20;               % SNR in dB


CFO = 0.05;             % Normalized CFO

phaseNoiseVar = 1e-3;   % Phase noise variance

iq_gain_imb = 0.1;      % IQ gain imbalance
```

```
iq_phase_imb = 5*pi/180; % IQ phase imbalance

%% TRANSMITTER

% QPSK Modulation
data = randi([0 3], N, numSymbols);
modData = pskmod(data, 4, pi/4);

% Insert pilots (every 4th subcarrier)
pilotIdx = 1:4:N;
modData(pilotIdx, :) = 1 + 1j;

% IFFT
tx_ifft = ifft(modData, N);

% Add CP
tx_cp = [tx_ifft(end-cpLen+1:end, :); tx_ifft];

txSignal = tx_cp(:);

%% CHANNEL IMPAIRMENTS

n = (0:length(txSignal)-1)';
```

```
% CFO effect
cfoSignal = txSignal .* exp(1j*2*pi*CFO*n/N);

% Phase Noise (Wiener process)
phaseNoise = cumsum(sqrt(phaseNoiseVar)*randn(size(n)));
pnSignal = cfoSignal .* exp(1j*phaseNoise);

% IQ Imbalance
I = real(pnSignal);
Q = imag(pnSignal);

I_new = (1 + iq_gain_imb) * I;
Q_new = (1 - iq_gain_imb) * Q;

iqSignal = I_new + 1j*Q_new.*exp(1j*iq_phase_imb);

% Add AWGN
rxSignal = awgn(iqSignal, SNR, 'measured');

%% RECEIVER

% Reshape
```

```
rxMatrix = reshape(rxSignal, N+cpLen, numSymbols);

% Remove CP
rx_noCP = rxMatrix(cpLen+1:end, :);

% FFT
rx_fft = fft(rx_noCP, N);

%% ------------------ ESTIMATION ------------------

%% 1. CFO ESTIMATION (Blind using CP correlation)
cfo_est = 0;
for i = 1:numSymbols
    sym = rxMatrix(:, i);
    cp = sym(1:cpLen);
    data_part = sym(N+1:N+cpLen);

    cfo_est = cfo_est + angle(sum(conj(cp).*data_part));
end
cfo_est = cfo_est / (2*pi*cpLen*numSymbols);

fprintf('Estimated CFO: %.4f\n', cfo_est);
```

```
% CFO Compensation
n2 = (0:length(rxSignal)-1)';
rx_cfo_corrected = rxSignal .* exp(-1j*2*pi*cfo_est*n2/N);

%% Reprocess after CFO correction
rxMatrix = reshape(rx_cfo_corrected, N+cpLen, numSymbols);
rx_noCP = rxMatrix(cpLen+1:end, :);
rx_fft = fft(rx_noCP, N);

%% 2. PHASE NOISE ESTIMATION (Pilot-Aided)
phase_est = zeros(1, numSymbols);

for i = 1:numSymbols
    rxPilots = rx_fft(pilotIdx, i);
    phase_est(i) = angle(mean(rxPilots));
end

% Phase correction
for i = 1:numSymbols
    rx_fft(:, i) = rx_fft(:, i) * exp(-1j*phase_est(i));
end

fprintf('Phase Noise Estimated & Corrected\n');
```

```
%% 3. IQ IMBALANCE ESTIMATION (Data-Aided)

% Simple estimation using image rejection
rx_conj = conj(flipud(rx_fft));
iq_est = mean(abs(rx_fft - rx_conj), 'all');

fprintf('Estimated IQ Imbalance metric: %.4f\n', iq_est);

%% DEMODULATION

demodData = pskdemod(rx_fft, 4, pi/4);

%% BER CALCULATION

% Ignore pilots
dataIdx = setdiff(1:N, pilotIdx);

txBits = data(dataIdx, :);
rxBits = demodData(dataIdx, :);

[numErr, ber] = biterr(txBits, rxBits);
```

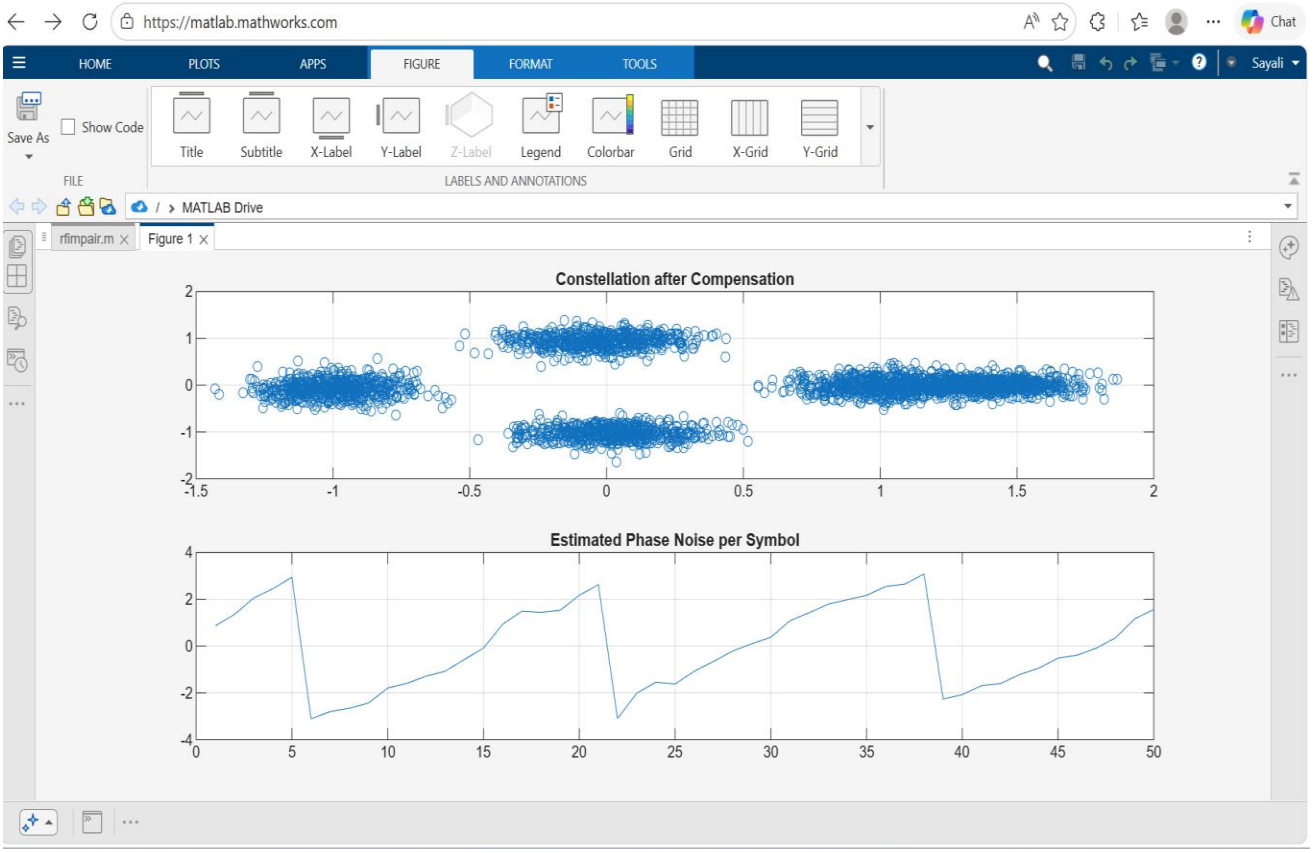
```
fprintf('BER: %e\n', ber);

%% PLOTS

figure;
subplot(2,1,1);
plot(real(rx_fft(:)), imag(rx_fft(:)), 'o');
title('Constellation after Compensation');
grid on;

subplot(2,1,2);
plot(phase_est);
title('Estimated Phase Noise per Symbol');
grid on;
```

result:-



Estimated CFO: 0.0031
Phase Noise Estimated & Corrected
Estimated IQ Imbalance metric: 1.3979
BER: 2.208333e-01